in Python. This software can be used for home automation and lighting control, though it is certainly not limited to these uses alone. It can work on any platform that supports Python (Windows, Mac OS X, Linux, etc).

***Main features*:**
- Executes actions based on voice inputs, time of day, file data, serial port data and socket data, as well as serial and speech protocols.
- REST API
- Mobile Web and Android clients with continuous device updates (Web sockets)
- Voice commands from Android ('Home Control' app)
- Local telnet and Web access
- Has a unique language to describe devices and action
- Works with smart objects, doors, lights, motion sensors, photo cells, etc
- Optional 'Mini loop' programming for more complicated controls
- Very easy to add new hardware drivers
- Good communication, complete with examples
- Maps one command to another with source and time
- Optional 'Event driven' programming, for complex actions when the device state changes

Freedomotic
This open source, flexible, secure Internet of Things (IOT) development framework is useful to build and manage modern smart spaces. It is integrated with popular building automation technologies like BTicino Open Web Net, KNX Modbus RTU and X10, as well as with custom automation projects using Arduino devices, Do It Yourself (DIY) boards, third party Android front-ends, text-to-speech (TTS) engines, motion detection sensors using IP cameras, social network integration like Twitter, and much more.

Freedomotic complies with well-known standards and building automation protocols as well as with 'do it yourself' solutions. It treats the Web, social networks and branded front-ends as first class components of the system. It can run on Raspberry Pi.

Main features:
- **Cross-platform:** Freedomotic is written in Java. So it can run on Windows, Linux, Mac and Solaris.
- **Distributed and scalable:** It can be deployed on a network of cheap peer-to-peer hardware nodes. It is scalable, and can be used in small apartments as well as large buildings.
- **Modular and extensible:** Freedomotic is modular, and can use plugins and cross-language APIs, which are distributed along with the software to easily create new add-ons.
- **Cross-language APIs:** You can connect different software, hardware, front-ends and services apps developed in your preferred programming language using REST, STOMP or plain Java.
- **Not a single front-end:** It can run many front-ends at the same time, even from a remote control.
- **Hardware friendly:** Freedomotic has an abstraction layer for hardware infrastructure (sensors and actuators).
- **Event based:** Every action in the real environment and every interaction with the system is mapped to an event.
- **Semantic (logic)-rich:** It provides a semantic-rich knowledge of environment topology, people and objects in order to implement intelligence and reasoning methods. No coding is required; the environment can be described using graphical editors.
- **History aware:** It can record environment changes in a separate and dedicated database. The historical series can be queried to provide charts, reports, alerts or even to be used to create history aware intelligent automations.
- **Multi-language:** Freedomotic is available in more than 20 languages.

Pi Dome
This home automation platform has been exclusively developed for Raspberry Pi. It provides solutions for end users as well as developers and hobbyists. Besides being a server, it also includes clients for multiple platforms.

Main features:
- Suitable for technical and non-technical users
- Runs multiple commands at the same time
- Has a dashboard for all client types
- MQTT broker with client functionalities
- Plugins for utility measurements, universal remote controls, SMSs, media (XBMC) and weather data
- Automatic creation of data graphs
- Allows you to write your own client with Java client libraries

Open source does offer a wide selection of automation solutions to help you make your home smart. You can research on this basic information and choose the solution you feel will work best for you. END

Reference
'Smart agricultural sensors' by Steven Schriber

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